

Web Technology



Revision 1 - HTML, CSS, Javascript

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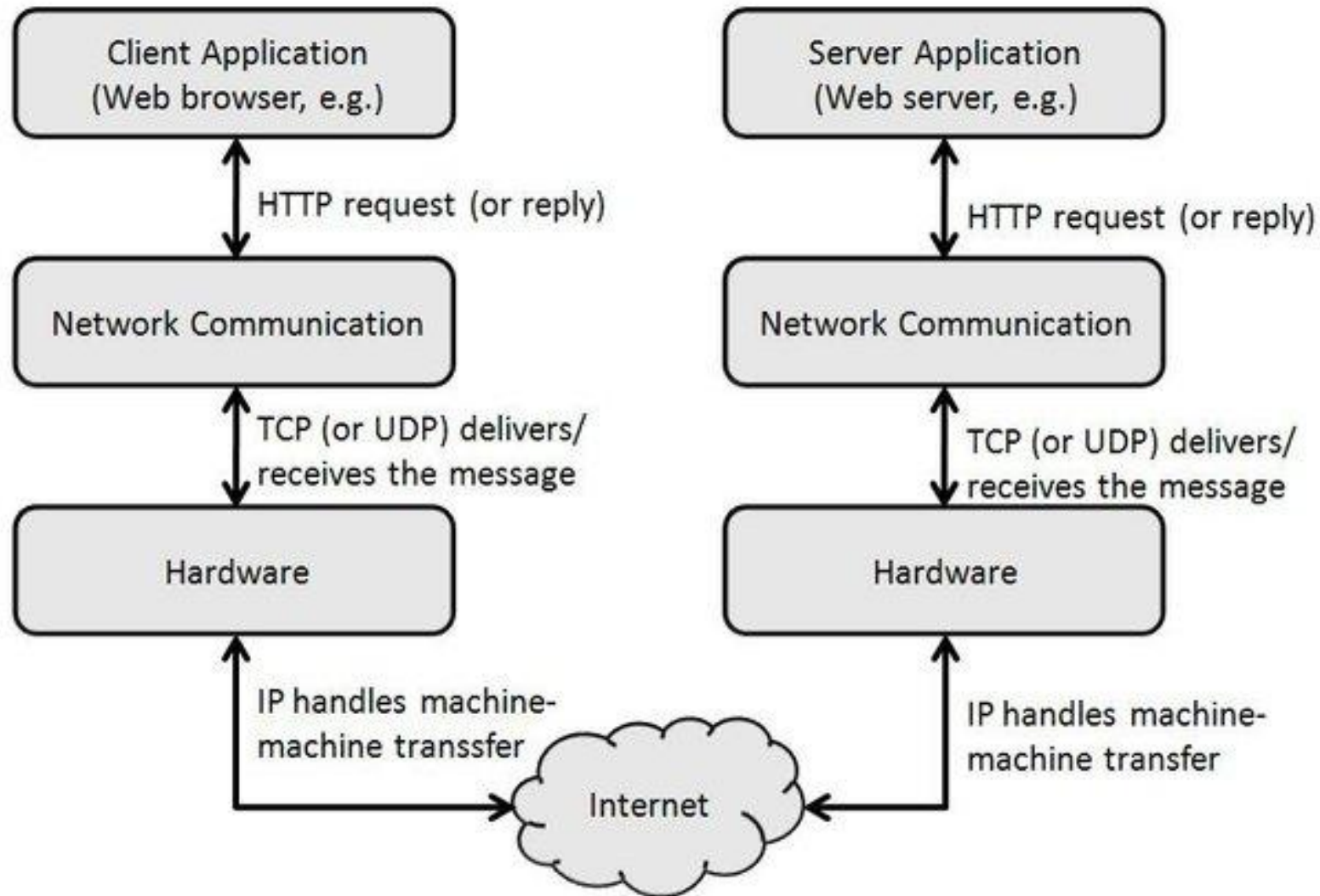
**Department of Computer Science and Engineering
Shiv Nadar University**

Software Development Life Cycle

6 Phases of the Software Development Life Cycle



Parts of the Web



HTML CSS Javascript



HTML

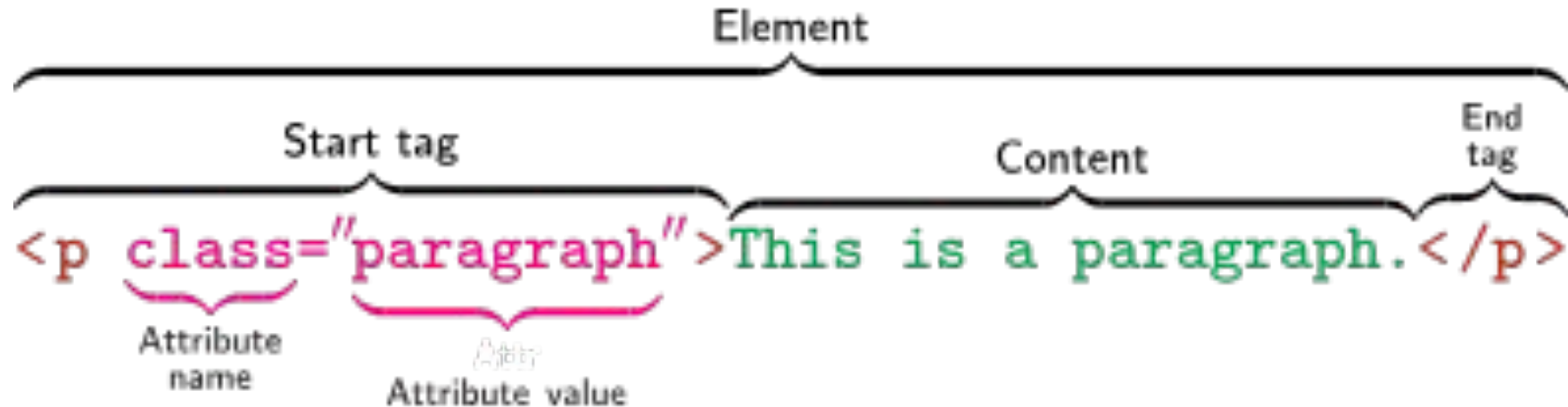


HTML + CSS



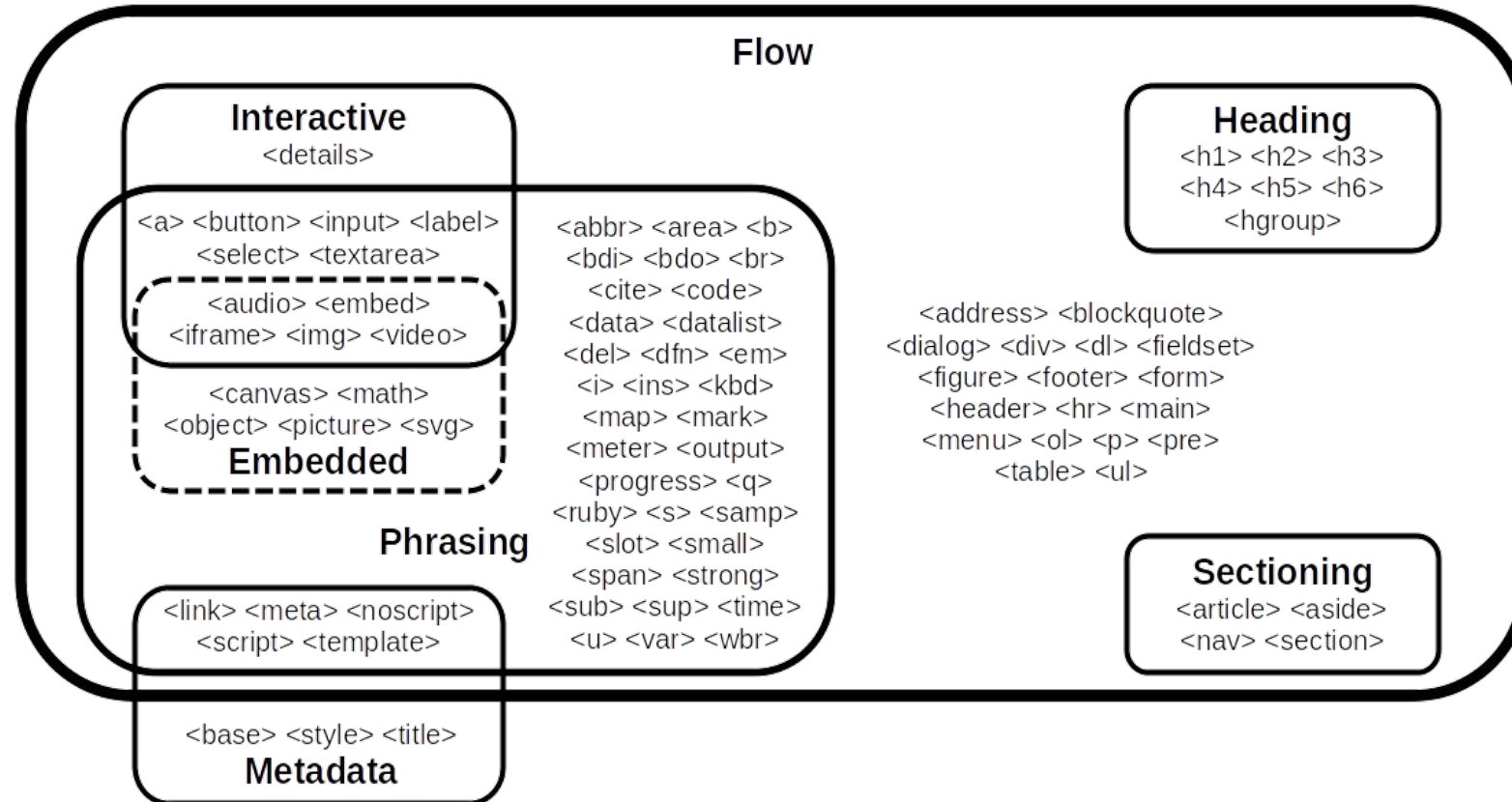
HTML + CSS + JavaScript

HTML Elements



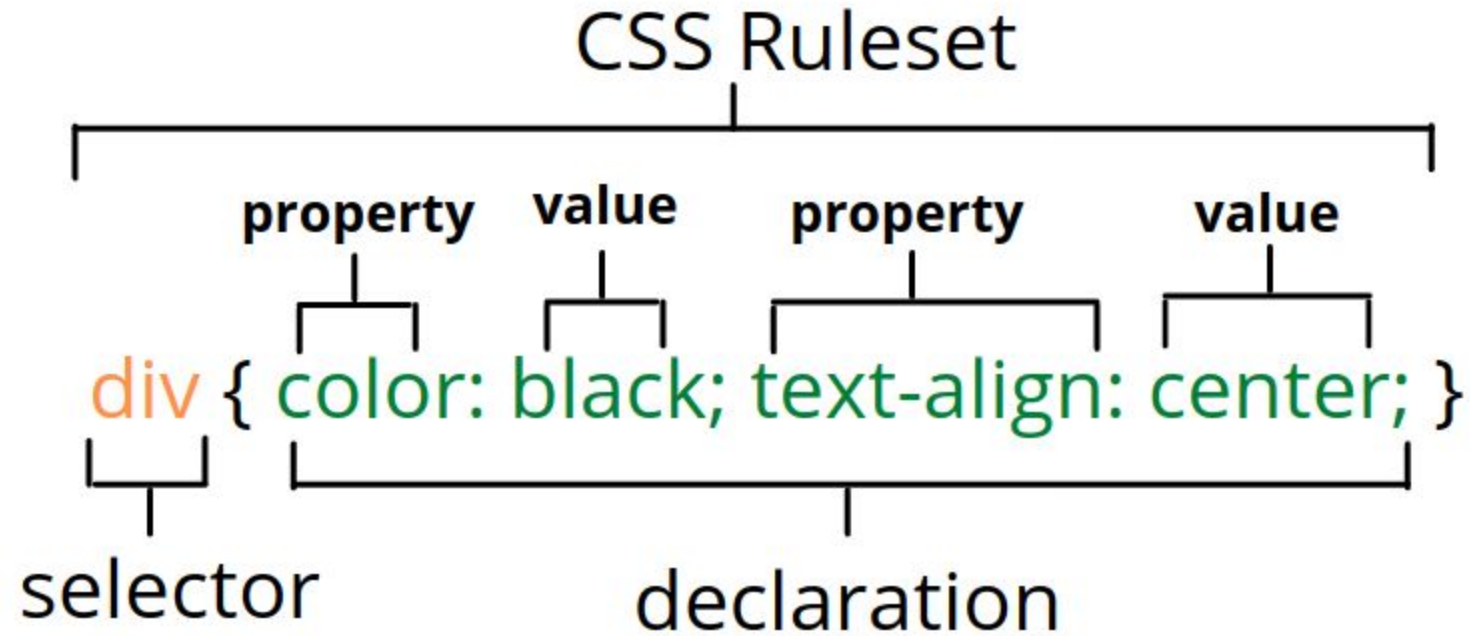
1. Elements – 110+
2. Attributes - 170

HTML element content categories



`<body>` `<caption>` `<col>` `<colgroup>` `<dd>` `<dt>` `<figcaption>` `<head>`
`<html>` `<legend>` `<li>` `<optgroup>` `<option>` `<rp>` `<rt>` `<source>`
`<summary>` `<tbody>` `<td>` `<tfoot>` `<th>` `<thead>` `<tr>` `<track>`

CSS Selectors



CSS Types

Inline CSS

```
<p style="color: blue;">This is a paragraph.</p>
```

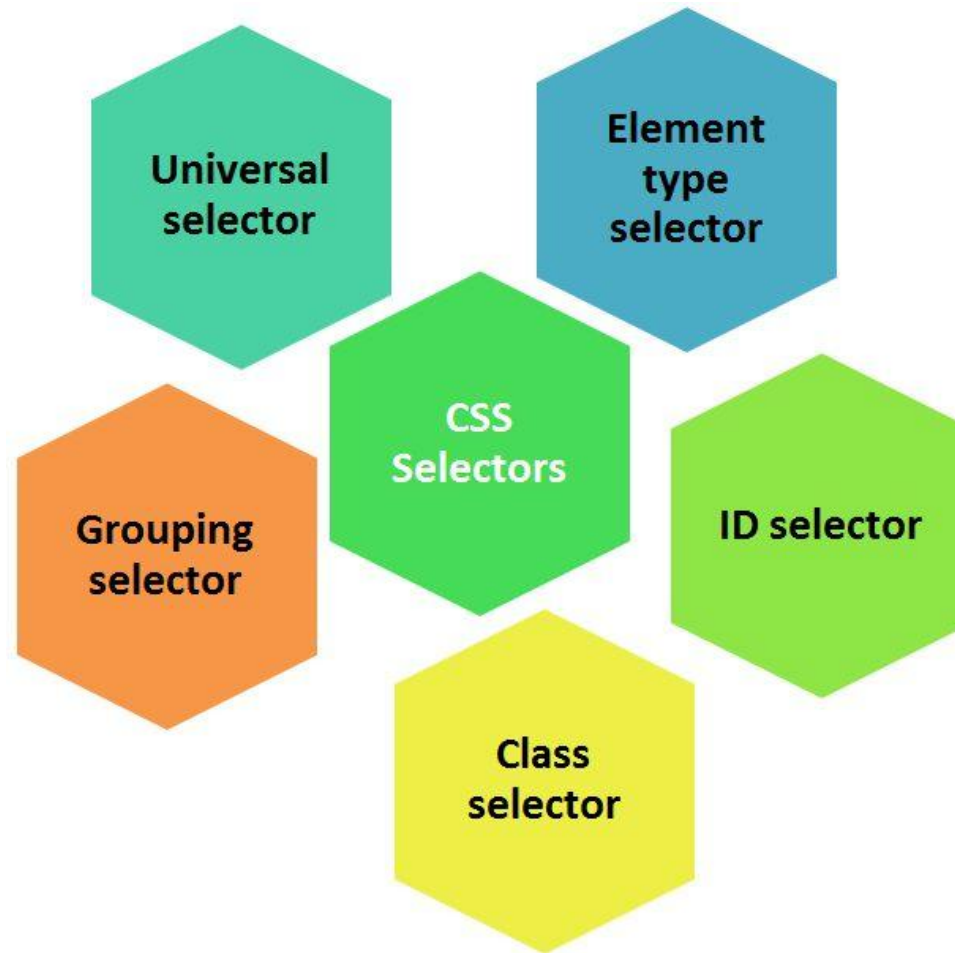
Internal CSS

```
<head>  
  <style type = text/css>  
    body {background-color: blue;}  
    p { color: yellow;}  
  </style>  
</head>
```

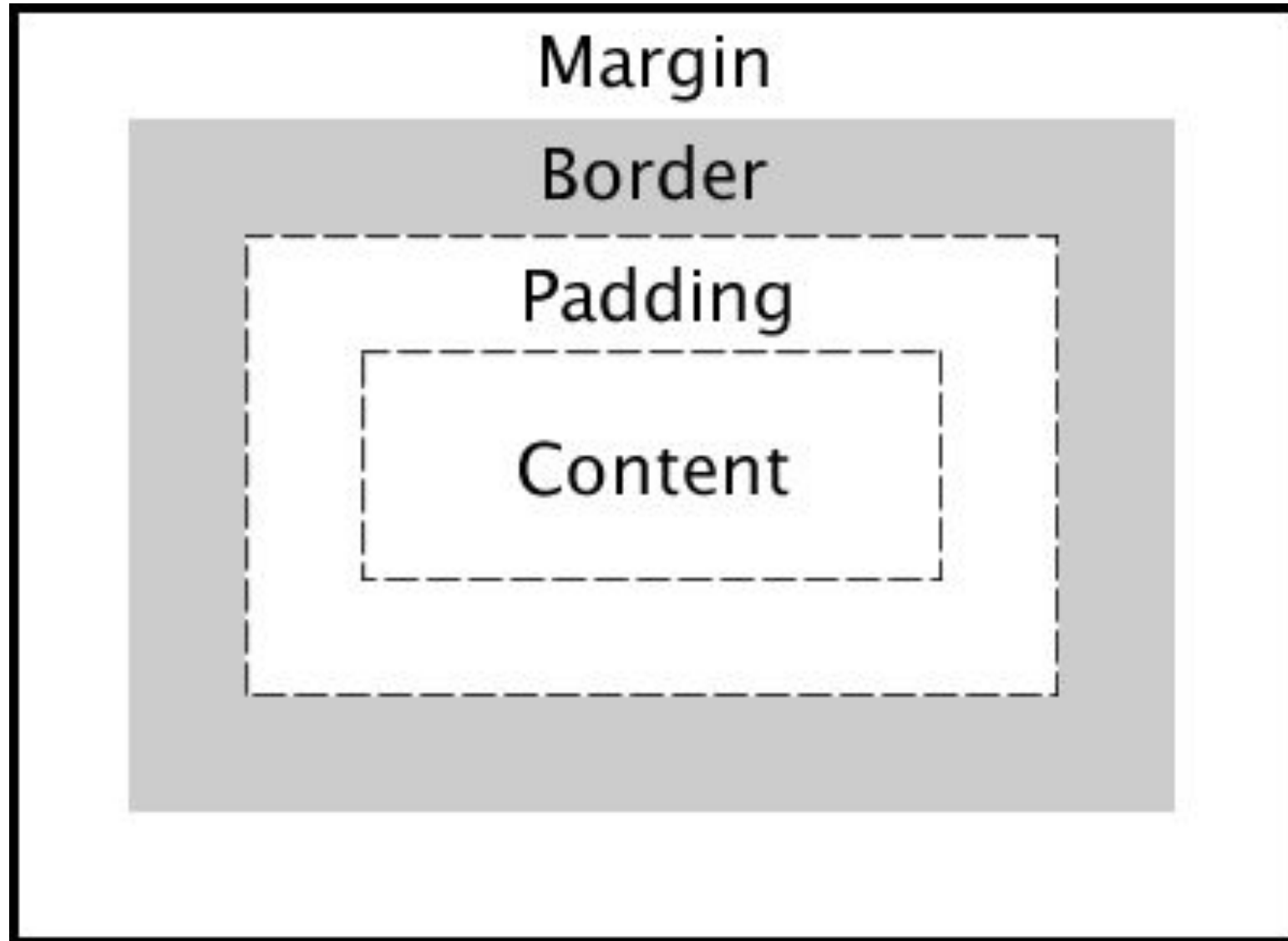
External CSS

```
<head>  
  <link rel="stylesheet" type="text/css" href="style.css">  
</head>
```


CSS Selector Types



CSS Box Model



var → **function-scoped (not block-scoped)**

```
if (true) { var x = 10; } console.log(x); // 10
```

x is available outside the if block.

let and const → **block-scoped**

```
if (true) { let y = 20; const z = 30; } console.log(y);  
// Error
```

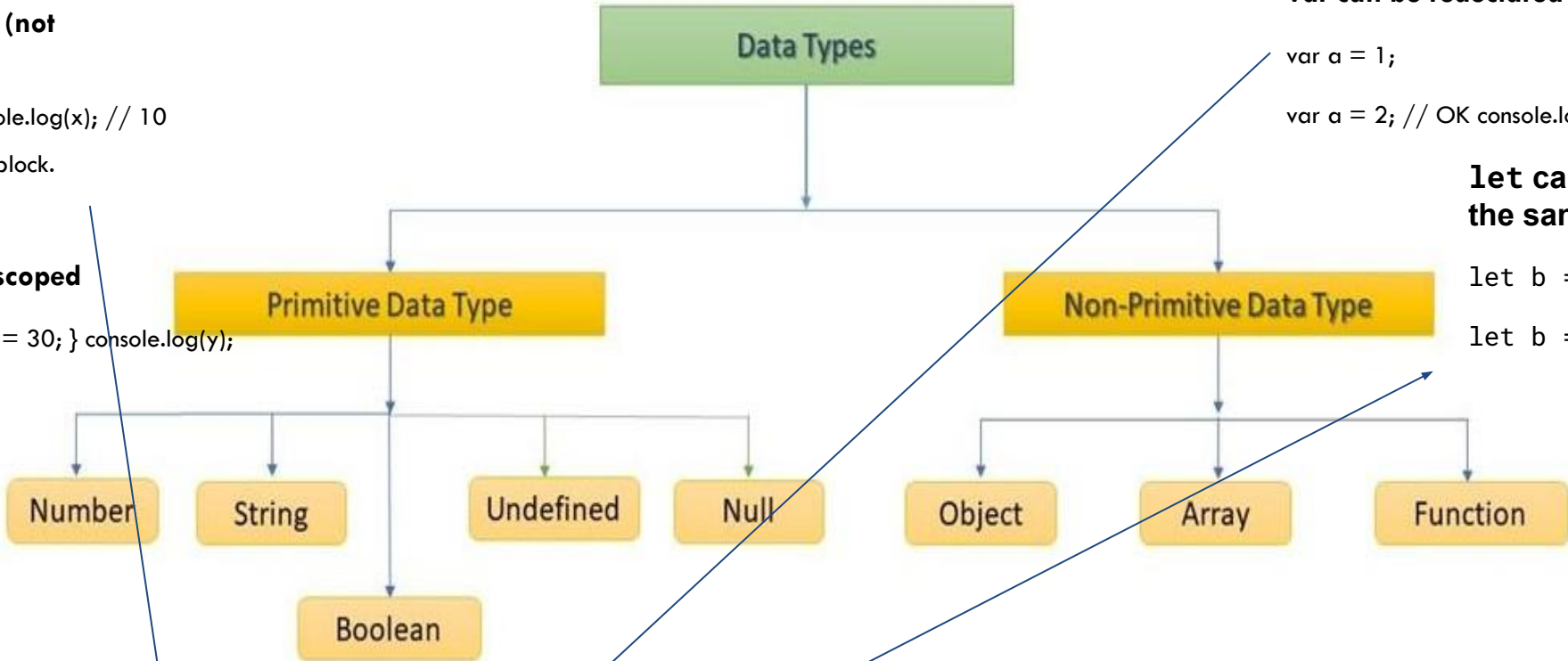
```
console.log(z); // Error
```

var can be redeclared

```
var a = 1;  
var a = 2; // OK console.log(a); // 2
```

let cannot be redeclared in the same scope

```
let b = 1;  
let b = 2; // Error
```



	Scope	Redeclare	Reassign	Hoisted	Binds this
var	No	Yes	Yes	Yes	Yes
let	Yes	No	Yes	No	No
const	Yes	No	No	No	No

Javascript

var

```
var a = 1;  
a = 5; // OK
```

let

```
let b = 1;  
b = 5; // OK
```

Hoisting

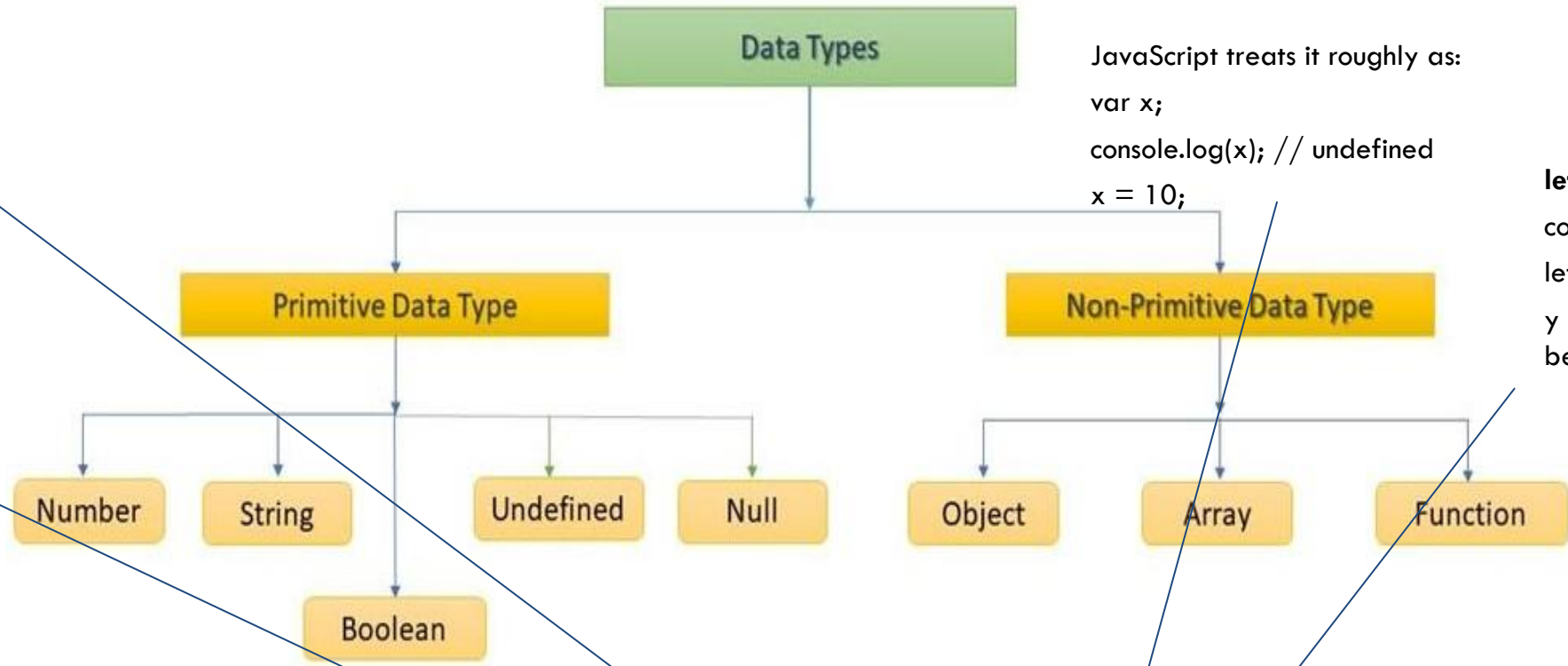
```
var  
console.log(x); // undefined  
var x = 10;
```

JavaScript treats it roughly as:

```
var x;  
console.log(x); // undefined  
x = 10;
```

let

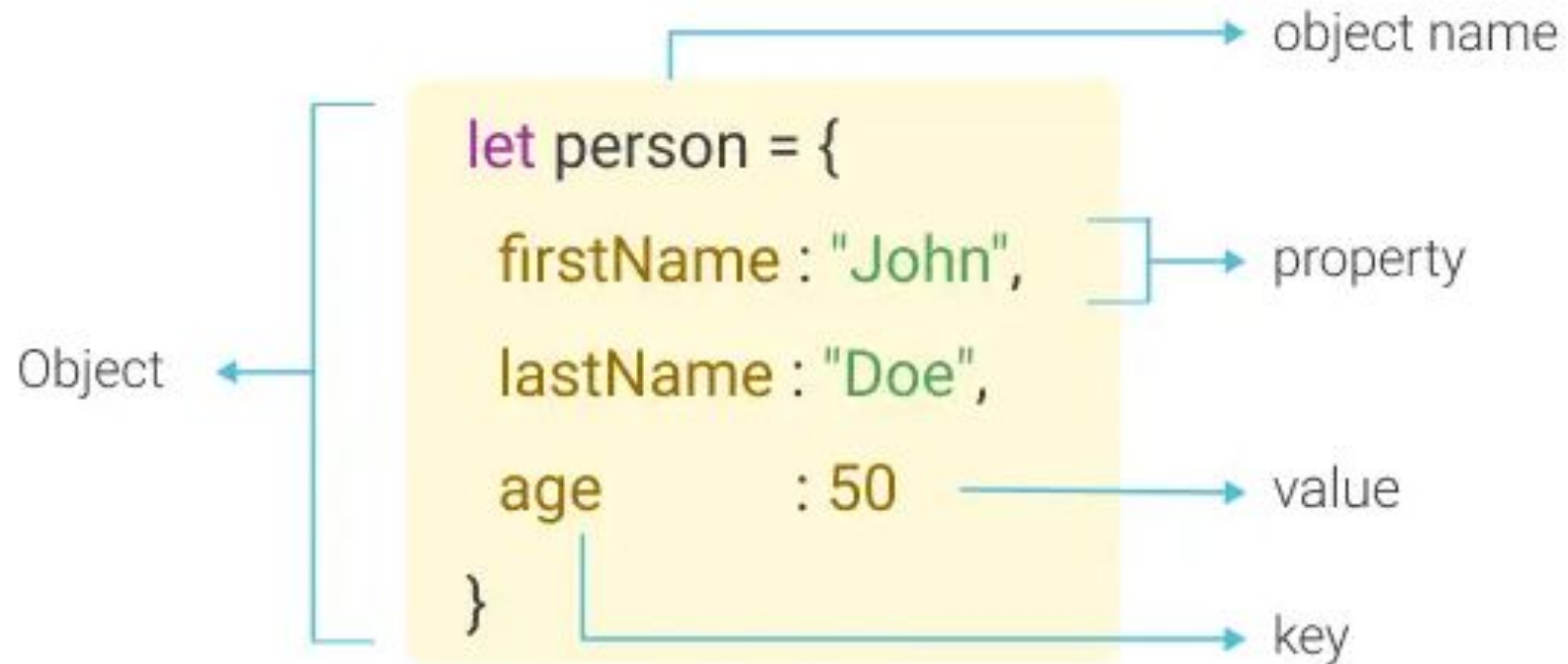
```
console.log(y); // Error  
let y = 20;  
y is hoisted but cannot be used  
before declaration.
```



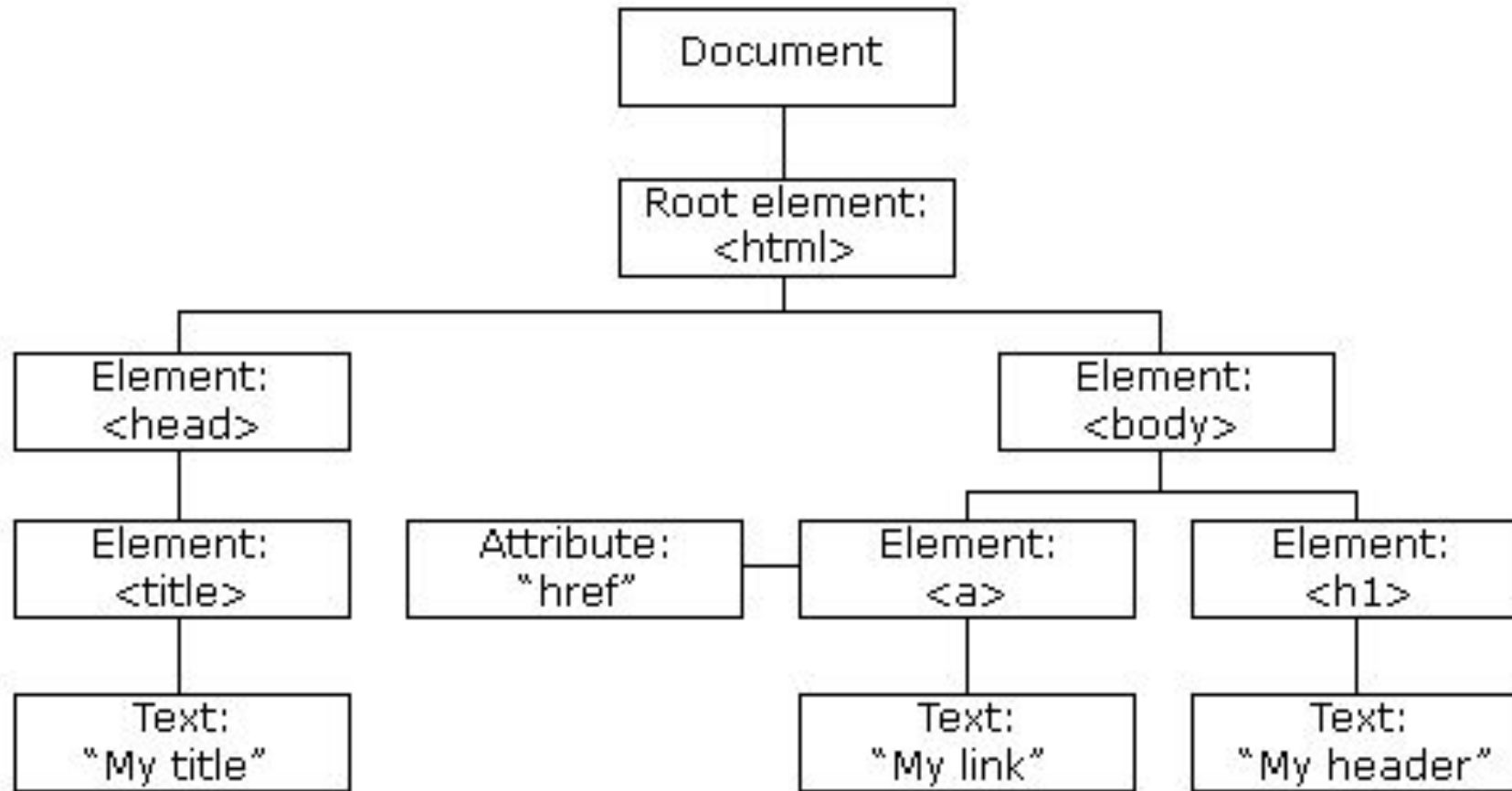
	Scope	Redeclare	Reassign	Hoisted	Binds this
var	No	Yes	Yes	Yes	Yes
let	Yes	No	Yes	No	No
const	Yes	No	No	No	No

Javascript - Objects, this

JavaScript Object Structure



Javascript - DOM



Course Outcomes

- **CO1 - Analyze and Build Dynamic Websites with HTML, CSS, Javascript, AJAX.**
- **CO2 - Analyze and Build Single Page Applications using React for Front-End Application Development.**
- **CO3 - Analyze and Build Node.js and Express for Back-End Development with Data Storage for Structured and Unstructured Data.**
- **CO4 - Analyze and Build MERN Stack for Full Stack Applications with RESTful APIs.**
- **CO5 - Apply Version Control using Git along with Project Deployment and Maintenance.**

1. **Paul J Deitel, Harvey M Deitel, and Abbey Deitel, "Internet and the world wide web: How to program," Pearson, Fifth Edition, 2012.**
2. **Brad Dayley, Brendan Dayley, Caleb Dayley , "Node.js, MongoDB and Angular Web Development: The definitive guide to using the MEAN stack to build web applications", Pearson Education, Second Edition, 2017.**
3. **Robin Wieruch, "The Road to React: The ReactJS with Hooks", Lean Publishing, 2023 Edition.**

THANK YOU

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